

U.S. Alternative Fuel Vehicle Analysis Report

Prepared for a Transportation Research Group and Government Transportation Board
Project focus: electric vehicle adoption, fuel-type distribution, and infrastructure investment priorities

Project Overview

This project analyzes the current distribution of electric, hybrid, traditional, and other alternative fuel vehicles across the United States using state-level vehicle registration counts. The main goal is to identify where electric vehicle adoption is strongest, where adoption is lagging, how EVs compare to gasoline and diesel vehicles, and which alternative fuels are significant versus niche. The results are intended to support government transportation planning, especially decisions about where EV charging infrastructure should be expanded and which markets automakers and policymakers should monitor most closely.

Dataset Summary

The raw dataset contains 51 rows and 13 columns. Each row represents a U.S. state or the District of Columbia, and each column records the number of registered vehicles by fuel type. The key fields are State, Electric (EV), Plug-In Hybrid Electric (PHEV), Hybrid Electric (HEV), Biodiesel, Ethanol/Flex (E85), Compressed Natural Gas (CNG), Propane, Hydrogen, Methanol, Gasoline, Diesel, and Unknown Fuel. The dataset is geographic and vehicle-registration based, so there are no demographic variables such as age, income, or population. Instead, the behavioral pattern captured in the dataset is the fuel-type mix in each state. For the final analysis, the District of Columbia was removed so the report focused strictly on the 50 states. Unknown Fuel was removed because it was not a specific fuel category, and Methanol was removed because its total was 0 across the dataset. After cleaning and feature engineering, the working dataset had 50 rows and 24 columns. There were no missing values and no duplicate rows.

Item	Rows / Count	Columns	Notes
Raw dataset	51	13	Includes 50 states plus District of Columbia
Final analysis dataset	50	24	District of Columbia, Unknown Fuel, and Methanol removed; calculated totals and percentages added
Missing values	0	-	No missing values found
Duplicate rows	0	-	No duplicate rows found

Exploratory Data Analysis and Cleaning using Python

Python was used first because the raw CSV contained vehicle count columns stored as text with commas, such as "1,256,600". Before doing analysis, those values had to be converted into numeric columns. The first EDA checks used `df.head()`, `df.info()`, `df.describe(include="all")`, `df.isnull().sum()`, and `df.duplicated().sum()` to understand the structure, identify data types, and confirm data quality.

```
import pandas as pd

df = pd.read_csv("Vehicle Data.csv")
df.head(10)
df.info()
```

```
df.isnull().sum()
df.duplicated().sum()
```

The District of Columbia row was removed to keep the analysis focused on the 50 states. Then all numeric fuel columns were converted from string/object values into integer values by removing commas and using `pd.to_numeric()`.

```
df = df.drop(index=8)

cols_to_convert = df.columns.drop("State")

df[cols_to_convert] = (
    df[cols_to_convert]
    .replace(",", "", regex=True)
    .apply(pd.to_numeric, errors="coerce")
    .astype("Int64")
)
```

Column names were standardized into snake_case to make them easier to use in Python, SQL, and Power BI. Unknown Fuel was removed because it was not useful for comparing specific fuel categories. Methanol was also removed because its sum was 0, meaning it did not contribute to the analysis.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.replace(r"\s*(.*)", "", regex=True)
df.columns = df.columns.str.replace(r"_$", "", regex=True)

df = df.drop(columns=["unknown_fuel"])
df["methanol"].sum()
df = df.drop(columns=["methanol"])
```

The most important feature engineering step was creating totals and percentage columns. Percentages were necessary because comparing raw vehicle counts alone would favor larger states. For example, California naturally has more EVs than smaller states, but EV percentage shows the actual adoption rate relative to each state's total vehicle base.

```
fuel_cols = df.columns.drop("state")
df["total"] = df[fuel_cols].sum(axis=1)

alt_fuel_cols = [
    "electric", "plug-in_hybrid_electric", "hybrid_electric",
    "biodiesel", "ethanol/flex", "compressed_natural_gas",
    "propane", "hydrogen"
]

df["alternative_fuel_total"] = df[alt_fuel_cols].sum(axis=1)
df["traditional_fuel_total"] = df[["gasoline", "diesel"]].sum(axis=1)
df["electrified_total"] = df[["electric", "plug-in_hybrid_electric",
    "hybrid_electric"]].sum(axis=1)

df["electric_percent"] = df["electric"] / df["total"] * 100
df["plug-in_hybrid_electric_percent"] = df["plug-in_hybrid_electric"] / df["total"] * 100
df["hybrid_electric_percent"] = df["hybrid_electric"] / df["total"] * 100
df["electrified_percent"] = df["electrified_total"] / df["total"] * 100
df["gasoline_percent"] = df["gasoline"] / df["total"] * 100
df["diesel_percent"] = df["diesel"] / df["total"] * 100
```

Key Python EDA Findings

The cleaned dataset contains 285,090,900 vehicles across the 50 states. Electric vehicles totaled 3,547,800, which equals 1.24% of all vehicles in the dataset. When EVs, PHEVs, and HEVs are combined, electrified vehicles totaled 12,225,500, or 4.29% of all vehicles. Gasoline remained the dominant fuel type at 85.10% of vehicles, and gasoline plus diesel together represented 87.62% of vehicles.

State	EV Count	EV %	Electrified %	Gasoline %
California	1,256,600	3.41%	9.15%	84.66%
Hawaii	25,600	2.38%	6.44%	88.35%
Washington	152,100	2.25%	7.39%	82.47%
Nevada	47,400	1.87%	5.03%	85.55%
New Jersey	134,800	1.85%	4.80%	88.27%

California had the highest EV adoption rate at 3.41%, followed by Hawaii at 2.38% and Washington at 2.25%. The lowest EV adoption rates were in North Dakota, Mississippi, Wyoming, West Virginia, and South Dakota, all below 0.20% EV share.

State	EV Count	EV %	Electrified %	Gasoline %
North Dakota	1,000	0.13%	1.27%	79.13%
Mississippi	3,600	0.13%	1.25%	83.82%
Wyoming	1,100	0.17%	1.61%	76.52%
West Virginia	2,800	0.19%	1.81%	85.68%
South Dakota	1,700	0.19%	1.65%	79.40%

Alternative Fuel Significance

Among alternative fuels, Ethanol/Flex (E85) was the most significant category. It represented 7.09% of all vehicles and was the dominant alternative fuel type in 49 out of 50 states. Hybrid Electric (HEV) vehicles were also meaningful at 2.59% of all vehicles, while EVs were 1.24% and PHEVs were 0.46%. Biodiesel was smaller nationally at 0.98%, but it was more important in some states such as Montana, Wyoming, North Dakota, South Dakota, and Idaho. Hydrogen, CNG, and propane were niche fuel categories. Hydrogen only appeared as a nonzero category in California, while CNG and propane represented extremely small shares nationally.

Fuel Type	Total Vehicles	Share of All Vehicles
Ethanol/Flex (E85)	20,223,800	7.09%
Hybrid Electric (HEV)	7,374,200	2.59%
Electric (EV)	3,547,800	1.24%
Biodiesel	2,803,200	0.98%
Plug-In Hybrid Electric (PHEV)	1,303,500	0.46%
Compressed Natural Gas (CNG)	24,600	0.01%
Hydrogen	16,900	0.01%
Propane	6,000	0.00%

Data Analysis using SQL (PostgreSQL)

After the cleaned DataFrame was created in Python, it was loaded into a PostgreSQL database table called `vehicle_analysis`. SQL was then used to answer the same business questions in a database environment. This step is important because it shows that the analysis can be reproduced outside the notebook and queried directly from a relational database, which is closer to how many production analytics workflows operate.

```
from sqlalchemy import create_engine
```

```
engine =
```

```
create_engine("postgresql+psycopg2://<username>:<password>@localhost:5432/usa_vehicle_analysis")
```

```
)
df.to_sql("vehicle_analysis", engine, if_exists="replace", index=False)
```

The first key SQL query ranked states by EV adoption rate. This directly answered which states had the highest EV adoption and which lagged behind.

```
SELECT
    state,
    electric,
    electric_percent,
    RANK() OVER (ORDER BY electric_percent DESC) AS electric_percent_rank
FROM vehicle_analysis
ORDER BY electric_percent DESC;
```

A second query ranked states by total electrified vehicle adoption, combining EV, PHEV, and HEV vehicles. This provides a broader view of the transition away from purely traditional fuel vehicles.

```
SELECT
    state,
    electrified_total,
    electrified_percent,
    RANK() OVER (ORDER BY electrified_percent DESC) AS electrified_rank
FROM vehicle_analysis
ORDER BY electrified_percent DESC;
```

The next SQL query compared alternative fuel dependence to traditional gasoline/diesel dependence and categorized each state. This made it easier to identify states that were further along in alternative fuel adoption versus states still strongly dependent on traditional fuels.

```
SELECT
    state,
    alternative_fuel_percent,
    traditional_fuel_percent,
    CASE
        WHEN alternative_fuel_percent >= 15 THEN 'High alternative fuel adoption'
        WHEN alternative_fuel_percent >= 12 THEN 'Moderate alternative fuel adoption'
        ELSE 'Low alternative fuel adoption'
    END AS alternative_fuel_category
FROM vehicle_analysis
ORDER BY alternative_fuel_percent DESC;
```

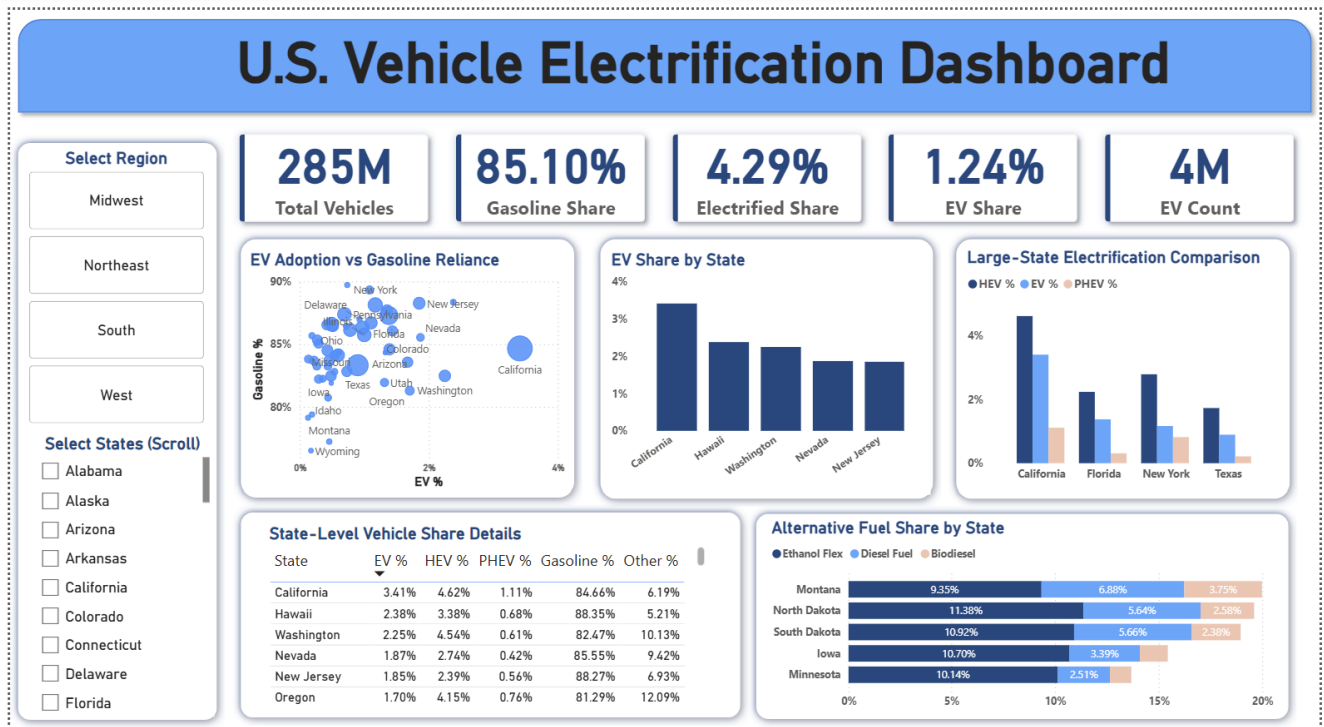
A grouping query was also used to identify the dominant alternative fuel type by state. This showed that Ethanol/Flex was the dominant alternative fuel in nearly every state, while California was the only state where Hybrid Electric was the dominant alternative fuel type.

```
SELECT
    dominant_alternative_fuel_type,
    COUNT(*) AS number_of_states
FROM vehicle_analysis
GROUP BY dominant_alternative_fuel_type
ORDER BY number_of_states DESC;
```

Overall, the SQL results supported the Python findings: EV adoption is strongest in California, Hawaii, Washington, Nevada, and New Jersey; weakest in North Dakota, Mississippi, Wyoming, West Virginia, and South Dakota; and traditional gasoline vehicles still dominate the national vehicle mix.

Dashboard in Power BI

Moving on from SQL, I connected Power BI to the vehicle_analysis database to import the data table. The Power BI dashboard was designed to make the analysis interactive. It includes KPI cards for total vehicles, gasoline percentage, electrified percentage, EV percentage, and EV total. It also includes Region and State slicers so the user can filter the dashboard. The main visuals include a scatter plot comparing EV share against gasoline share, a bar chart ranking states by EV adoption, a clustered column chart comparing EV, PHEV, and HEV percentages, a state-level table, and an alternative-fuel comparison chart. The dashboard makes it easy to identify high-adoption states, lagging states, and states where traditional fuel dependence remains high.



Business Recommendations

Based on the analysis, EV infrastructure investment should be prioritized in California, Washington, and Florida.

State	EV Count	Total Vehicles	EV %	Electrified %	Gasoline %
California	1,256,600	36,842,400	3.41%	9.15%	84.66%
Washington	152,100	6,769,400	2.25%	7.39%	82.47%
Florida	254,900	18,487,100	1.38%	3.93%	87.29%

California: California should be the first priority because it has both the highest EV count and the highest EV adoption rate. It has 1,256,600 EVs, an EV share of 3.41%, and an electrified vehicle share of 9.15%. Even with this leadership, gasoline still represents 84.66% of vehicles, so the state needs continued public charging expansion, fast-charging capacity, grid support, and charging access for apartment and urban residents.

Washington: Washington should be the second priority because it combines high adoption with meaningful scale. It has 152,100 EVs, an EV share of 2.25%, and an electrified share of 7.39%. Since Washington is already one of the strongest EV adoption states, additional charging infrastructure would help prevent charger shortages, support highway travel, and expand adoption beyond the strongest metro areas.

Florida: Florida should be the third priority because it has the second-largest EV count in the dataset at 254,900 EVs, but its EV share is only 1.38% and gasoline still represents 87.29% of vehicles. This combination suggests a large market with growing EV demand but substantial room for adoption growth. Charging investment in Florida would support current EV drivers while also helping convert a very large gasoline-dependent vehicle market.

Additional recommendation: lagging states such as North Dakota, Mississippi, Wyoming, West Virginia, and South Dakota should not be ignored, but their immediate need may be more focused on incentives, education, corridor planning, and baseline charging access rather than large-scale deployment. These states have very low EV shares, so policymakers should first identify barriers such as rural travel distance, vehicle affordability, charging availability, and consumer awareness.

Conclusion

The analysis shows that the U.S. vehicle market remains heavily gasoline-dependent, but EV and electrified vehicle adoption is growing unevenly across states. California is the clear leader, while Washington, Hawaii, Nevada, New Jersey, Oregon, Colorado, Maryland, Arizona, and Florida also show relatively strong EV adoption. Ethanol/Flex is the most widespread alternative fuel category, while hydrogen, CNG, and propane remain niche. For infrastructure planning, the strongest strategy is to invest first in states where EV demand is already visible and where large gasoline-dependent vehicle bases create major conversion potential.